

Measuring the Liberal Arts in the Age of Big Data:
Lessons and Opportunities from Computational Social Science

Annaliese Paulson

University of Michigan

I. Introduction

Historically, quantitatively measuring a liberal arts education as scale has been difficult because large-scale administrative datasets and traditional social science surveys do not typically capture the core constructs that define a liberal arts education. A liberal arts education might be characterized by educational practices like intensive writing and regular faculty-student interactions, educational content such as encounters with the “Great Books of the Western World” or breadth of study, and aspects of the educational environment like student interactions on residential campuses. However, variables for these constructs are not readily available in the typical quantitative dataset.

Rapid transformations in data infrastructure, increases in computational capacity, and the increasingly prominent role of the internet and computers in mediating educational processes have transformed the kinds of data available to social science researchers. These transformations allow researchers to observe behavior and ask questions that were previously inaccessible to quantitative social scientists. To name a few of the new forms of data from this digital transformation, student’s academic experiences over time are now captured in rich longitudinal administrative student transcripts; course content, assigned readings, and pedagogical practices are posted online in the text of course catalogs and syllabi; student interactions with faculty and peers are recorded through email and discussion boards; exams, papers, and quizzes are recorded in learning management systems; and student interactions with their peers on campus can be inferred through WiFi routers and friendships observed on social media websites. Drawing on the recently emerged field of computational social science, this book chapter provides guiding principles for those seeking to use digital data to measure aspects of a liberal arts education and explores the opportunities the digital revolution provides for measuring constructs related a

contemporary liberal arts education. To preview what follows, in Section II, I discuss the challenge of quantitatively measuring a liberal arts education. In III, I introduce principles of computational social science and discuss the strengths and weaknesses of digital trace data. In IV, I discuss examples of digital trace data that can be used to quantitatively measure the liberal arts, drawing on existing literature using novel data sources, and, in V, I conclude by discussing the challenges and opportunities of building an interdisciplinary research agenda that uses digital trace data to measure liberal arts education.

II. The Challenge of Measuring a Liberal Arts Education

Although the terms “liberal education” and “liberal arts education” are widely used and most colleges in the United States argue they provide some form of a liberal education, there is substantial ambiguity in the meaning of the terms. A liberal arts education could be defined by the kind of institution a student attends, the design of the course of study, or the educational aims or objectives of the course of study (Brighouse, 2019). According to Brighouse, the simplest definition of a liberal arts education is made by appealing to the kind of institution a student attends: a liberal arts education is an education that takes place in a small liberal arts college or in a school of letters, arts, and sciences within a larger institution. This definition is lacking, however, in mechanisms that link attending a liberal arts school to its educational outcomes.

A second more nuanced approach to defining the liberal arts Brighouse offers appeals to the content of a student’s course of study: a liberal education typically includes both breadth of study through electives and general education requirements and depth in a major. On this interpretation, an education could be more or less aligned with liberal education; we might say students that balance breadth and depth receive a liberal education while students that achieve

too much depth with too little breadth are pursuing a specialized course of study that is less aligned with the liberal arts.

In addition to breadth and depth of study identified by Brighouse, we can add the educational context of a liberal arts education, following Detweiler (2021). Detweiler argues a liberal arts education is typically characterized by engaging pedagogy and active learning, an emphasis on writing coursework, small class sizes, active discussion in courses, and living on or near a residential campus. While the educational practices of a liberal arts education may be more common at liberal arts institutions, they are not the exclusive domain of those schools (Pascarella et al., 2005; Detweiler, 2021). For instance, to some, liberal arts education and the approximately third of a curriculum devoted to general education are synonymous (Pascarella et al., 2005; Wells, 2016) and all accrediting agencies require postsecondary institutions implement some form of general education coursework (Wells, 2016). For those who equate general education to liberal arts education, all institutions provide some degree of a liberal arts education. However, simply understanding the processes associated with a liberal arts education may be insufficient. While liberal arts education is characterized by distinct processes, understanding these processes does not inform whether they are effective.

Finally, consider Brighouse's third approach to defining a liberal arts education: appeals to the aims and objectives of the liberal arts. For instance, in 2007, the Association of American Colleges and Universities (AAC&U), a prominent advocate of the liberal arts, offered the followings aims of a liberal arts education in *College Learning for a New Global Century*:

Knowledge of Human Cultures and the Physical and Natural World

Intellectual and Practical Skills, Including: Inquiry and analysis; Critical and creative thinking; Written and oral communication; Quantitative literacy; Information literacy; Teamwork and problem solving

Personal and Social Responsibility, Including: Civic knowledge and engagement—local and global; Intercultural knowledge and competence; Ethical reasoning and action; Foundations and skills for lifelong learning

Integrative and Applied Learning, Including: Synthesis and advanced accomplishment across general and specialized studies (p. 13)

On this definition, a liberal arts education is any education that achieves these aims. Achieving these outcomes may be facilitated by attending liberal arts institutions as in Brighouse's first definition or by experiencing the content and practices that define his second definition.

However, there is no necessary logical link between attending such an institution or experiencing particular educational practices and these outcomes. Ultimately, it is an empirical question how the practices of a liberal arts education are linked to the outcomes of such an education.

The plurality of ways one might define a liberal arts education and the fact that they are not necessarily linked with one another presents a measurement problem for quantitative social scientists interested in studying how liberal arts education affects students. We want to understand the mechanisms through which a liberal arts education - broadly defined - affects a wide range of student outcomes. Table 1 presents a non-exhaustive list of the diverse processes and outcomes that have been associated with a liberal arts education. Many of these proposed components of a liberal arts education are difficult to measure quantitatively in traditional social science surveys: there are not, for example, longitudinal measures of the assignments and course content that develop writing, oral communication, and quantitative reasoning available in

surveys. Measurement challenges like this one have motivated the development of the interdisciplinary field of computational social science.

Table 1

Theoretical Constructs Related to the Processes and Outcomes of a Liberal Arts Education

Theoretical Construct	Process or Outcome	Source
Student-Faculty Engagement	Process	Detweiler (2021) Pascarella et al. (2005)
Engagement with Great Books	Process	Bok (2006), Montás (2021) Wells (2015)
Nonvocational Coursework	Process	Detweiler (2021)
Writing Assignments	Process	Detweiler (2021)
Active Learning and Engaging Pedagogy	Process	Campbell (2023) Detweiler (2021)
Residential Living	Process	Delbanco (2012) Detweiler (2021) Brighouse (2019)
Breadth of Study	Process	AAC&U (2007) Bok (2006) Detweiler (2021) Brighouse (2019)
Participation in Student Organizations or Athletics	Process	Detweiler (2021)
Engagement with Diverse Perspective	Process	Bok (2006) Detweiler (2022) Pascarella et al. (2005)
High Impact Practices	Process	Kuh (2008)
Cognitive Competencies and Dispositions	Outcome	AAC&U (2007) Wells (2016) Delbanco (2012) Detweiler (2021)

General Skills/Human Capital	Outcome	AAC&U (2007) Pascarella et al. (2005) Delbanco (2012) Wells (2016) Wildasky (2023)
Democratic Engagement	Outcome	AAC&U (2007) Delbanco (2012) Wells (2016) Detweiler (2021)
Appreciation for the Arts	Outcome	Detweiler (2021)
Moral Character and Ethical Reasoning	Outcome	AAC&U (2007) Bok (2006) Wells (2016) Pascarella et al. (2005)

III. Computational Social Science and Data in the Digital Age

Responding to the digital transformation of society, the recently emerged interdisciplinary field of computational social science draws on statistics, computer science, data science, and the traditional social sciences to expand the range of social science questions that can be asked and answered (Edelmann et al., 2020; Salganik, 2018). Because computational social science works with data that were not designed for social science research, it requires a combination of computer science to develop tools that work with new forms of data and social science to understand how digital data can be meaningfully mapped onto existing theories of human behavior (Lazer & Radford, 2017).

Historically, quantitative social science research has been conducted with large scale surveys from organizations like the National Center for Education Statistics and the Bureau of Labor Statistics. These surveys are designed for research purposes and include measures for a wide range of theoretical constructs. However, because they are costly and must appeal to a broad range of researchers, traditional surveys are limited in a number of ways.

First, because administering high-quality surveys is labor and time-intensive, the sample sizes were relatively small, and surveys were not often administered frequently enough to capture development over time (Salganik, 2018). Second, surveys seldom capture an individual and all of their peers, limiting researchers' ability to study relational network structures (Macy, 2016). Finally, because surveys must cover a wide range of theoretical constructs, the measures are, by necessity, broad; researchers interested in relationships between theoretical constructs need to find the best proxy for their constructs in the survey and, if no proxy is present, there is no remedy (Grimmer et al., 2022).

The rise of digital infrastructure has transformed the data that quantitative social scientists can access. Students' educational lives are now mediated through what Matt Salganik (2018) calls *digital traces*, or digital records of processes recorded in digital databases. Because digital traces are produced as a record of actions, they typically record actual behavior, rather than the self-reports used in surveys (Lazer & Radford, 2017). Rather than asking a student to report how many times they access course materials online, digital traces record every time a student actually accessed those materials. For example, digital traces include novel sources of data like call records, social media data, the digitized text of documents like newspapers and syllabi, and videos posted online that have not frequently been used in quantitative social science research. However, Salganik also argues that many forms of administrative data that have become commonplace in quantitative social science should also be considered digital traces and share many of the same strengths and limitations. Administrative data from Unemployment Insurance records, student transcripts, and voting records are intentionally designed as a record of actions individuals have taken and have not been designed with researchers' aims in mind. In combination, novel sources of data like YouTube videos and digitized newspapers and the vast

digital databased from governments and businesses comprise the digital traces that computational social science studies.

As educational processes are increasingly mediated through digital technology and computers (Fischer et al., 2020), social scientists can use the digital traces left behind by those processes to quantitatively measure theoretical constructs at a scale and granularity that was previously impossible. To name a few digital traces from education: course-taking and grades are recorded in digital administrative transcripts, exams and papers are submitted through online learning management systems, course catalogs and syllabi describing the content students learn are posted online, and web logs capture the time students spend accessing education resources online. In contrast to traditional quantitative surveys, these data come in a variety of non-tabular formats including video, images, text, and networks and often come at a scale that makes them inaccessible to traditional quantitative methods without significant cleaning and pre-processing (Lazer & Radford, 2017).

While digital traces offer timely and always-on measures of student's behaviors at tremendous scale, they present distinct challenges for social scientist as they were not designed for research. As a result, digital traces must be thoughtfully repurposed to be used as valid measures of theoretical constructs. By carefully understanding the process that generates digital trace data from student actions, the digital transformation of educational processes can enable the quantitative study of the processes and outcomes of a liberal arts education. In the following sections, I discuss ten facets of digital trace data identified by Salganik (2018), focusing first on the three virtues of digital trace data for social science and second on seven limitations of using this data in research. I encourage interested readers to refer to Matt Salganik's *Bit by Bit: Social Research in the Digital Age* and the 2020 review of computational social science by Edelman et

al. for further discussion of the strengths and weakness of digital trace data in social science research.

Three Virtues of Digital Trace Data: Big, Always-On, and Nonreactive

Big. In contrast to traditional surveys, digital trace data are often big because they passively capture entire social behaviors (Lazer & Radford, 2017). For instance, the administrative student transcripts from four-year institutions in Texas comprise more than 50 million total enrollments between 2011 and 2022. Although the size of a dataset is not a virtue in and of itself, big data can enable the studying of rare events including more credible causal inference designs like regression discontinuities and allow for the analysis of heterogeneity of effects across different student identities (Card et al., 2010; Figlio et al., 2017; Lazer & Radford 2017; Penner & Dodge, 2019).¹ Further, because digital trace data often capture entire populations, they allow researchers to construct the relational social networks of individuals, rather than observing single students in isolation (Macy, 2016). For instance, because administrative student transcript data typically captures the universe of students at an institution rather than a sample, these digital traces allow researchers to construct the full social context of peers in an individual's coursework.

Understanding heterogeneity of effects is particularly important given the longstanding interest and debate about which aspects, if any, from the historical (and white European) liberal arts tradition benefit students from identities outside of that historical tradition. W.E.B. Du Bois (1903/2004; 1933/2004) and trailblazing Black feminist Anna Julia Cooper (1930/2022), for example, both discussed whether Black Americans should develop vocational skills or pursue a liberal arts education to most effectively navigate a social and economic world defined by white

¹ Currie et al. (2020) and Goldsmith-Pinkham (2024) provide empirical evidence of the increasing reliance on digital trace data in empirical economics.

supremacy. However, quantitatively modelling heterogeneity across identity groups requires sample sizes large enough to include interactions between quantitative measures of liberal arts education and measures of identity. Given the cost of large-scale surveys, it is not always possible to ensure sufficient sample sizes (Figlio et al., 2017; Viano & Baker, 2020), making it difficult to quantitatively understand how aspects of a liberal arts education might interact with identities without turning to big digital trace data. Relative to traditional social science surveys, the size of big data offers numerous benefits for those studying the liberal arts. It allows for the quantitative study of rare events and more credible causal inference, for modelling heterogeneity of effects across demographic groups, and allows for the modelling of the relational network structure of educational processes.

Always-on. In contrast to surveys that are administered at specific time-points, digital trace data are often always-on, passively recording behaviors at all times. This always-on nature is one of the primary reasons digital traces are also typically big. For instance, both the Postsecondary Education Transcript Studies associated with NCES surveys like the Beginning Postsecondary Survey 12:17 and a typical university administrative database collect student transcripts. However, the BPS contains a snapshot of just one cohort's course-taking collected for research purposes at great expense. In contrast, because administrative student transcripts are used as a record of student course-taking, they are regularly updated each academic term and contain all student transcripts since the establishment of the administrative database. Researchers can access transcripts from decades of student cohorts rather than just one.

Because digital trace data are always-on passively recording behavior, independent of changes in the environment (Lazer & Radford, 2017), digital traces like administrative Unemployment Insurance data and student transcripts provide data before and after policy

changes that facilitate causal inference through difference-in-differences designs (Figlio et al., 2017; Salganik, 2018). Further, digital traces can also facilitate research on the long-term consequences of policies and allow researchers to study development over time (Penner & Dodge, 2019; Edelman et al. 2020). For example, because digital traces of employment records are recorded in always-on Unemployment Insurance databases, researchers have traced the evolution of labor market outcomes of liberal arts and vocational majors over decades (Andrews et al., 2022).

However, the always-on nature of digital trace data can be overstated. Without intentional archiving, digital traces from social media and institutional websites may disappear over time. An institution is incentivized to maintain digital data as long as it continues to serve an administrative purpose. Current information may be kept up to date but links to information from previous years may not be maintained. While we can typically find a course catalog for the 2023 academic year online, the probability we can find a comparable document for the 2005 academic year is much lower. Despite the best efforts of organizations like the Internet Archive, online digital trace data is not eternal. For instance, the Pew Research Center found that almost a quarter of links that were posted between 2013 and 2023 are no longer available with 38 percent of links posted in 2013 being unavailable (Rivero et al., 2024). Although digital trace data is always-on, simply being on is no guarantee that it will stay available if it ceases to serve its intended purpose.

Nonreactive. Measurement using digital trace data is less likely to change behavior in response to that measurement. In traditional social science measurement, there is a concern that asking survey questions may change respondent's behavior because they know they are being observed, known as reactivity. If researchers regularly ask students how much time they spend

studying, they may respond by increasing their studying because they perceive being a good student as socially desirable and are under regular observation. In contrast, if we measure how much time students spend studying through alternative mechanisms – for example the number of days they interact with online course materials – they are less likely to consciously change their behavior. As a result, of passive recording of behavior, this makes digital traces nonreactive.

These first three characteristics of digital trace data — big, always-on, and nonreactive — are advantageous for social science research. When contrasted with traditional social science surveys, digital trace data often has the scale to allow for the measurement of rare events and effect heterogeneity, allows for causal identification designs that exploit changes over time and allows for studies that follow evolution over time because it is always-on, and does not change behavior as a result of measurement. As a result, digital trace data have notable benefits over traditional quantitative social science surveys. However, the remaining characteristics of digital trace data present challenges for the development of meaningful social science measures.

Seven Vices of Digital Trace Data

Salganik (2018) identifies seven characteristics of digital traces that make them challenging to use: digital trace data are incomplete; are inaccessible to many researchers; contain sensitive information; are not representative of a well-defined sample population; experience drift over time; are often algorithmically confounded; and are dirty, requiring substantial cleaning. In this section, I discuss each of these limitations of digital trace data in turn.

Incomplete. In contrast to surveys, digital trace data often lack clear measures of theoretical constructs. For instance, the Faculty Survey of Student Engagement contains self-reported measures of the emphasis faculty place on writing clearly and effectively (Nelson Laird

et al., 2009). However, no comparable measure of writing is available in administrative student transcripts: we typically observe only student course-taking and grades.

Administrative data like student transcripts are designed for administrative purposes with data recording course-taking and grades as marks of student academic progress. Since there is limited value to administrators in measuring a course's writing emphasis, it is not routinely recorded in these data sources. Social scientists seeking to use administrative data quickly realize that their data is incomplete. Indeed, some of the most common demographic variables from traditional social science surveys in education — measures of race, gender, and socio-economic status — are often missing from digital trace data (Macy, 2016) and, even when present, demographic variables like race/ethnicity are often measured at a much coarser level than theory would suggest is useful for understanding the social world (for example, collapsing the diversity of Asian students lived experiences into one demographic category, Viano & Baker, 2020).

Because digital trace data are incomplete, researchers are forced to identify creative proxies for constructs or focus on research questions with easily measured theoretical constructs. This incompleteness shapes the research possible with digital trace data; it is easier to examine some outcomes of a liberal arts education like labor market outcomes with ready proxies captured in state Unemployment Insurance datasets or aspects of democratic engagement proxied by voting records. It is harder to imagine deriving a credible measure of appreciation of the arts or a love of life-long learning from digital traces.

Inaccessible. In general, digital trace data are difficult to access because they are held by organizations that do not routinely make their data publicly available to researchers (Fischer et al. 2020; Salganik, 2018). As a result, accessing data from social media companies, third-party education vendors, or postsecondary institutions typically requires active partnerships and is – in

many cases – conditional on the status of the researcher (Edelmann et al., 2020). Consider – for example – that the social media website LinkedIn captures large-scale economic data that links college graduates to jobs, firms, and locations over time. Although research partnerships with LinkedIn allow for novel tests of longstanding social science theories (for instance, Rajkumar et al., 2022), LinkedIn data is inaccessible to most researchers. Webscraping LinkedIn data is against terms of service and the company actively attempts to prevent webscraping through legal action (*hiQ Labs, Inc. v. LinkedIn Corp.*, 2022). Although some researchers have leveraged LinkedIn data without partnerships (for example, Conzelmann et al., 2023), universities may be unwilling to approve research using this data given outstanding legal questions about its use or a desire to maintain positive relationships with Microsoft, LinkedIn’s parent organization. Although some researchers have begun purchasing this data from third party vendors (for example, Amornsiripanitch et al., 2023), rich digital traces from LinkedIn that track liberal arts graduates career growth and geographic mobility at scale exists but is inaccessible to most researchers.

In addition to restricting data access, inaccessibility means research questions can be constrained because access is conditional on an external partner; organizations are more likely to approve research that benefits the organization and are unlikely to allow research that portrays them unfavorably (Salganik, 2018). Further, because digital trace data are inaccessible, Stopczynski et al. (2014), argue that they are locked in “silos.” When accessing a single form of digital trace data requires building and maintaining relationships, it becomes difficult to construct datasets that capture all the theoretically linked constructs. Although email exchanges, WiFi routers, and call records each capture an aspect of a student’s social network, studies often rely on a single channel, resulting in an incomplete picture of social networks.

Sensitive. Data from many educational institutions, governments, and organizations are sensitive and protected. This is particularly true in education where most digital traces of student behaviors at postsecondary institutions are protected by the Family Educational Rights and Privacy Act: identifiable data on students can only be shared with researchers to develop, validate, or administer predictive tests; administer student aid programs; or improve instruction. Because of these constraints, many of the most valuable digital traces of educational processes are particularly inaccessible. Although procedures to securely access sensitive data have been developed (Card et al. 2010), restrictions put in place due to its sensitivity make democratic access to digital trace data challenging and require substantial investment in research infrastructure to facilitate access while respecting the data’s sensitivity (Penner & Dodge, 2019; Edelman et al., 2020).

Nonrepresentative. Unlike surveys, digital trace data are not typically drawn from a larger population with a well-defined sampling frame. Digital traces are not created for research purposes, so they are not designed to be representative of any sample population. Because digital traces often provide a census of some population and are big, they can provide the illusion of representativeness (Lazer & Radford, 2017). This results in what Lazer et al. (2014) call “big data hubris”: the belief that the volume of data can solve problems of generalizability that traditional quantitative surveys are carefully designed to overcome.² Without a well-defined sampling frame however, it can be difficult to know exactly what population the census represents, and researchers typically lack knowledge of the mechanisms of inclusion and selection that create populations in digital trace data. As a result, nonrepresentative digital traces

² Even less charitably, D’Ignazio and Klein (2020) characterize this tendency to valorize big data as “Big Dick Data”, connecting big data projects that ignore context, fetishize size, and inflate technical capabilities to patriarchal fantasies of world-domination through data capture and analysis.

have limited ability for out-of-sample generalizations although within-sample comparisons can still provide novel insights. Foote and Stange (2022) offer a helpful illustration of the challenges of nonrepresentative data. Researchers interested in studying labor market returns increasingly turn to state administrative Unemployment Insurance records because they offer always-on, passive collection of wage data; however, because the data are limited to recording employment within one state, Unemployment Insurance data fail to capture students who migrate out of state after their education and may misstate the true effect of educational experiences.

Drifting. Because digital trace data are designed for administrative use rather than research purposes, the content of these data systems can change over time. Salganik (2018) notes at least three ways this occurs: population drift, behavioral drift, and system drift. Population drift occurs when the individuals using a system change over time, behavioral drift occurs when the way individuals use the system change, and system drift changes when the system itself changes. Because digital trace data can drift over time, it may be challenging to derive consistent measures of theoretical constructs that enable meaningful comparisons over time. While being always-on makes long term follow up possible, the drifting nature of this data means we must validate that the population, behavior, and system have not meaningfully changed before making these comparisons.

Algorithmically Confounded. In many digital systems, digital trace data are not naturally occurring because they are generated, in part, by the engineering goals of the system. This is particularly true in social networks like Facebook and LinkedIn, which algorithmically make recommendations for content to follow and connections to create. Social networks on online social networks do not just reflect student's actual social ties, they create them. For example, Overgoor et al. (2020) use Facebook to map the online social networks of 1,159 public

institutions in the United States and show how different types of institutions have different social network structure. While interesting in its own right as Facebook is a major social media network that affects student lives, we cannot immediately make inferences from the structure of Facebook social networks to student's full social networks. In part, the structure of the student's online social network is the product of the algorithms that makes recommendations to them, not just their natural behavior.

Dirty. Big data sources are dirty, containing a significant amount of data that do not reflect the real actions and theoretical constructs of interest to researchers. Consider, for instance, using the text of a syllabus as a measure; following Arum and Roksa (2011), we might study whether enrolling in courses with substantial writing is associated with liberal arts majors or applied majors, using the text descriptions of required assignments as a proxy for whether the course requires a substantial amount of writing. While syllabi typically include a short description of evaluation methods and assignments, they also include significant amounts of text that would be extraneous to this research question like administrative details about the course and mental health resources. To use syllabi in this way, we must map the dirty text into a meaningful social science measure. We cannot, for example, infer from the presence of words like "write" in the syllabus that a student is required to write an essay – the presence of the word could as easily refer to requirements for students to write emails about questions or write computer code as part of a programming assignment. Because digital trace data is dirty, it often requires more work to clean and validate measures that capture the theoretical construct of interest.

Digital Trace Measures Require Validation

By passively recording the everyday interactions individuals have with digital systems, digital trace data have the potential to capture student behavior at scale over time in a nonreactive manner. However, these data are messy and not designed for research purposes. To use these digital traces as social science measures, we must create a mapping between the theoretical measure and the observed digital trace data that results in construct validity, a match between data and theoretical constructs.

Given that digital traces record actual behavior, it can be tempting to view measures derived from them as more objective than surveys that rely on self-reports. Digital trace data often have the allure of objectivity because behaviors are represented as quantitative values that provide an appearance of mathematical objectivity (D'Ignazio & Klein, 2020), appear to capture the entire universe of a population of interest (Lazer & Redford, 2017; D'Ignazio & Klein, 2020), and record actual behavior rather than self-reports (Lazer & Redford, 2017). However, simply being a quantitative measure of a behavior recorded for administrative purposes does not guarantee that a measure captures a meaningful theoretical construct. Salgaik (2018) argues that problems of construct validity are even greater for digital trace data than when using traditional survey measures because we are often repurposing imperfect dirty and incomplete proxies of theoretical constructs. To use digital trace data, social science researchers must first find a way to quantitatively map the rich complexity of the world present in imperfect digital trace data to a small number of quantitative values that measure a theoretical construct of interest construct (Grimmer et al., 2022). They must then validate their measure by building an argument for the construct validity of their measure. A successful argument will show how the derived measure behaves appropriately in relation to other available measures of similar constructs – often derived from surveys – and theoretically derived hypotheses about how the measure should

relate (Grimmer et al., 2022). When imagining what could produce a strong argument for construct validity, contrasting an idealized measure with the measure derived from digital trace data provides a helpful illumination of the often-strong assumptions necessary to repurpose digital traces for social sciences (Salganik, 2018). To that end, Margolin et al. (2013) encourage researchers to ask, “what does [survey response x] look like in [behavioral data y]?” in an effort to translate digital traces into meaningful social scientific measures (p. 9). Without thorough validation of measures derived from digital traces, we must be skeptical of making strong theoretical conclusions on the basis of this data.

IV. The Measurement Possibilities of Digital Trace Data

In this section, I survey aspects of the liberal arts that can be measured with digital trace data, presenting a non-exhaustive list of novel digital traces that have been used in computational social science. Once researchers recognize that the digital transformation of educational processes has resulted in passive always-on collection of data, many digital processes become possible social scientific measures from WiFi routers (Nguyen et al., 2020) to library website search records (Kabo et al., 2024). I present the examples below as inspiration for researchers curious how they can use the increasingly digital world to observe the processes and outcomes of a liberal arts education.

Course Expectations and Assignments.

A liberal arts education is associated with an emphasis on intensive writing assignments and de-emphasis on rote recitation of knowledge through quizzes and tests (Detweiler, 2021). Further, the AAC&U have argued for the educational benefits of several kinds of assignments that they have identified as high impact practices including collaborative assignments and projects, service or community-based learning, and writing-intensive coursework (Kuh, 2008;

Kilgo et al., 2015). Measuring the assignments and evaluation criteria students encounter at scale can provide insight into the mechanisms through which a liberal arts education operates.

Measuring Course Expectations and Assignments with Learning Management Systems. In the digital era, an increasing amount of a student's course content and assignments are mediated through learning management systems (LMS) like Canvas and Blackboard. In many courses, digital traces of student assessments like essays and online exams are now captured in online databases, proxies of course content are available on course pages that contain journal articles and other course readings, and syllabi are hosted on landing pages (Lewis et al., 2005; Fischer et al., 2020). Using the data present in an LMS, researchers can measure whether students are required to submit writing assignments and complete quizzes or exams.

Further, for courses that do require students submit assignments through a LMS, we can examine aspects of these assignments. Crossley et al. (2023), for example, use natural language processing techniques to examine essay writers' ability to integrate information using source material by deriving theoretically grounded quantitative measures of writing from the text of 909 essays. In doing so, they demonstrate the feasibility of using natural language processing to evaluate student essays at scale. For course sections that require essay submissions through learning management systems, researchers can extract student essays from learning management systems and use tools like those developed by Crossley et al. to derive quantitative measures of student writing at scale. One could, for example, identify all courses that meet university general education writing requirements, extract the submitted essays from the learning management systems for those courses, and use natural language processing to derive theoretically interesting measures of student's writing. When arranged longitudinally, this provides a portrait of student writing over time using passively recorded always-on data.

However, properly navigating the affordances of scale in LMS pose potential measurement problems. Faculty are not required to use these systems: they might require assignments be submitted through alternative means or link to courses readings rather than post them online. Without validation, we cannot infer from the absence of learning materials that these behaviors did not occur. Further, learning management systems change over time, introducing new features and removing old ones, resulting in system drift. If the changes are large enough, assessing longitudinal trends becomes difficult as the digital trace data we are using to create measures changes over time. Finally, comparing behavior across learning management systems can be challenging: two institutions may use LMS's in different ways and provide different functionality (Fischer et al. 2020). As a result, measures derived from the digital traces of a learning management system at one institution may not be valid measures at another. While these measurement challenges can be surmounted through thorough validation and knowledge about the specific institutional context a LMS is used in, they present challenges to meaningfully using LMS data at a scale greater than a single institution.

Measuring Course Expectations and Assignments with Syllabi. Syllabi contain rich information about coursework – typically including a description of the course content, a list of required readings, and a list of assignments and evaluation criteria – and are increasingly available at scale. Most notably, the Open Syllabus Project maintains a large-scale corpus of more than 18.7 million syllabi. The Open Syllabus Project corpus is constructed through webscraping: automated scripts crawl institution websites, classify documents they find as to whether they resemble syllabi, and add them to a corpus that has been made available to researchers (for instance, Biasi & Ma, 2022; Börner et al., 2018; Chau et al., 2023). Individual institutions or schools also increasingly maintain syllabi archives to help students navigate the

curriculum, several states have passed laws requiring faculty post syllabi online (for example, Texas House Bill 2504 in 2009 and Florida Senate Bill 7044 in 2022), and syllabi can be extracted from learning management systems for research purposes (Jiang & Pardos, 2020).

Assuming that syllabi contain a faithful representation of course assignments, researchers can derive measures of course assignments from syllabi that can be linked with student outcomes. Biasi and Ma (2022) identify the sections of syllabi that relate to course evaluation using natural language processing techniques then create measures of what they call “soft-skills intensity,” following Deming (2017). They measure soft skills by identifying a number of words describing assignments identified as high impact practices – for example “group”, “team”, and “essay” – while they measure hard skills by counting the number of words referring to “homework”, “exam”, and “quiz”. A syllabus has soft-skills intensity if it has more words associated with soft skills than hard skills. They find that their measure of soft-skills intensity is higher in liberal arts schools, in schools with higher expenditure on instruction and salaries, in more selective schools, in schools whose students have a higher median parental income, and in schools with a lower share of minority students. Examining the relationship with student outcomes, a one standard deviation increase in soft-skills intensity is associated with a 1.2 percentage point higher graduation rate, 1.7 percent higher earnings, and a 1.2 percentage point increase in the probability of reaching the top quintile of earnings among students whose parents were in the bottom quintile of earnings. Note, however, that their measures derived from syllabi cannot capture what students do – the syllabus only captures what an instructor intended students complete at the point of time the syllabus was posted. Students do not always follow the curricular plans laid out for them and instructors may change assignments over time without updating the course syllabi.

Course Content

For many, a liberal arts education is defined by a difference in course content. For some, this means teaching what are referred to as the Great Books of the Western World (Bok, 2006; Montás, 2021). For others, it involves an explicitly nonvocational study and breadth of study (Detweiler et al., 2021). Drawing on assigned textbooks, course titles, and course syllabi, digital trace data offers many proxy measures for the content that defines the liberal arts tradition.

Measuring Course Content with Assigned Textbooks. A prominent strand of liberal arts education has argued for the study of “great” books, or the set of books academics believe are the finest examples of human writing and thinking (Bok, 2006; Montás, 2021). Often - though not always – the list of books is drawn exclusively from the Western canon including, for example, Plato, Aristotle, Augustine, and Descartes. Over the past ninety years there have been dramatic debates over the place of the Great Books in the liberal arts tradition with accusations that universities have “abandon[ed] universal dignity” by removing these books from requirements (Sasse, 2023) contrasted with claims that advocates of the Great Books are embracing fascism (Stanley, 2020). These debates over the importance of the great books are, as Bok observed in 2006, largely devoid of empirical with just one survey of freshman from 1985 quantitatively examining the relationship between the Great Books and student success (Astin, 1993).

Since 2008, the Higher Education Opportunity Act has required each institution’s online course schedule include information about required textbooks including ISBN numbers, authors, titles, and the cost of textbooks. The digital traces left by this legislation in institutional websites can allow researchers to examine whether reading the Great Books truly affects students by linking the textbooks a student is assigned to administrative transcripts.

Beyond measuring exposure to specific books, digital trace data from textbooks offers proxies for content by linking textbook ISBN information with external databases like the Library of Congress and WorldCat. Using these external databases, researchers can link ISBNs to a description of each book provided by the publisher and a list of short text descriptions about the content a book is associated with (for example “calculus”, “educational sociology” or “ethics”). Observing that a required textbook is associated with history, math, or business categories provides a potential proxy for a course’s focus on the nonvocational content associated with the liberal arts or is more vocationally oriented.

Like syllabi, books are limited by the fact that we only observe whether they are assigned, not whether they are read and engaged with. Knowing that a student is assigned Plato’s *Republic* does not guarantee they have engaged with its arguments. Without careful validation work that shows how assigned textbooks relate to what students do, we can only use assigned books as a measure of what students are asked to do, not what they actually do in their coursework.

Further, assigned textbooks provide an incomplete picture of course content. Many instructors use a mix of textbooks and journal articles, and this will differ across disciplinary contexts: a math course may have a single textbook that covers its content, while an economics course could include a combination of journal articles and books. In the former case, we can argue that measuring the textbooks assigned provides a good summary of the course content, while in the later we have an incomplete picture.

Measuring Course Content with Course Titles. Administrative student transcripts often include a string course title, for example “Intro to Molecular Biol”. While remarkably short and often abbreviated, course titles can have sufficient information to identify some forms of

content. For instance, Zhou (2023) uses the short text of administrative transcripts to identify experiential learning and internship courses – two forms of course taking recommended as high impact practices by the AAC&U (Kuh, 2008). After controlling for observables using propensity score matching, they find a positive association between enrolling in experiential learning courses and the probability of employment after graduating and wages two and three years after graduating among four-year degree recipients but a negative association between experiential learning coursework and wages among two-year degree recipients.

Similarly, Borisova (2018) and Paulson et al. (2024) both use natural language processing techniques with supervised machine learning to classify short course titles into taxonomies of courses, identifying whether a course title is predicted to belong to broad disciplines like mathematics or economics, and fine-grained categories like calculus or geometry. For instance, in Paulson et al. (2024), we use labor-intensive human annotated transcripts from four waves of the National Center for Education Statistics' Postsecondary Education Transcript Studies as training data for a supervised machine learning that accurately classifies the short titles present in administrative course transcripts into one of 1,622 fine-grained courses. We use these codes as measures of the disciplines that student's course-taking from 19 institutions is drawn from, showing the majority of undergraduate course-taking comes from the traditional liberal arts disciplines – the humanities, social sciences, natural sciences, and fine arts. Although applied disciplines are now the majority of majority of majors at the end of our sample, the liberal arts continue to define the academic core of course taking.

Labor Market Demand for Skills.

Advocates for the liberal arts now frequently make appeals to the economics value of a liberal arts education. Wildavsky (2023), for example, argues that the contemporary economy

requires a combination of broad general skills – critical thinking, the ability to work in teams, and effective written and oral communication – and specific technical skills. Such arguments for the economic value of a liberal arts education rest on the premise that the broad skills developed by the liberal arts are in demand in the labor market and that technical skills face eventual obsolescence. A newly developed vein of research using digital trace data from online job boards and digitized newspaper have begun providing the empirical backing for these claims.

Measuring Labor Market Demand for Skills Using Job Advertisements. Online job boards now provide always-on data on the changing skill demands of jobs. For example, using online job advertisements from the labor market analytics firm Lightcast (formerly Burning Glass), Deming and Noray (2020) examine the changing skill content of job ads between 2007 and 2019. Lightcast curates the near-universe of online job ads, uses natural language processing techniques to map the text of job ads into binary measures of more than 15,000 skill requirements, and classifies each job advertisement into a six digit Standard Occupation Code. Using this data, Deming and Noray find significant change in skills over the period in question: many skills that are in demand in 2019 did not exist in 2007 and many skills required in 2007 are obsolete by 2019. Technology intensive fields – STEM occupations and business fields like advertising, logistics, and market research – have the highest rate of skill change, consistent with the claim that technology intensive and more vocationally oriented skills are at greatest risk of obsolescence.

Connecting their measures of skill change with survey data on college majors and labor market outcomes from the American Community Survey (ACS), they show that graduates from more applied fields work in occupations with greater skill change than majors in broader fields. Because rapidly changing occupations require more on the job learning to keep up with changing

skill demand, they develop a simple model that predicts a flatter age-earning profile in rapidly changing occupations and results in an incentive to switch out of technology intensive occupations over time. They corroborate their model predictions by testing them against ACS data. In combination, their measures of skill change over time from jobs advertisements, theoretical model, and survey results, suggest that narrow technical skills are at greater risk of obsolescence, providing a theoretical and empirical justification to claims about the value of broad skills and a liberal arts education.

Similarly, Deming and Kahn (2018) use Lightcast data to examine labor market returns to types of skills in professional occupations. After controlling for occupation, they find that variation in demand for social skills like teamwork and cognitive skills like problem solving have a positive association with measures of pay and firm performance. They also find evidence of a complementary relationship between social and cognitive skills, suggesting that social skills may magnify the effect of cognitive skills. Demand for cognitive and social skills positively predicts variation in wages, suggesting that the general skills a liberal arts education is associated with are valued in the labor market. Using the same Lightcast data, Hemelt et al. (2023) examine how employer skill demand differs across majors. They similarly find that demand for customer service and financial skills are restricted to a limited number of majors while social and organizational skills are demanded across most majors.

Atalay et al., (2020) draw on an alternative dataset of job advertisements but reach similar conclusions about the importance of cognitive skills and a shift away from routine cognitive and manual tasks. Drawing on a longitudinal dataset of digitized newspapers from the Boston Globe, New York Times, and Wall Street Journal between 1950 and 2000, they extract 7.8 million jobs from the scanned text of newspapers and classify each job advertisement into a

Standard Occupation Code using machine learning models. They validate their measures by comparing them against changes in widely used measures of tasks in jobs and by replicating key findings from previously conducted research on trends in occupation task changes. Using this novel dataset, they then show that jobs have shifted away from routine cognitive and manual tasks toward nonroutine cognitive tasks that require more critical thinking and interactive tasks between individuals. A substantial share of this change occurs within occupations, suggesting the work force is increasingly shifting toward nonroutine tasks that require critical thinking.

Digital trace data from job advertisements offer a convincing argument for the value of the broad skills associated with the liberal arts. Specific technical skills face obsolescence in a rapidly evolving economy (Deming & Noray, 2020), there is evidence of an increasing return to cognitive and social skills (Deming & Kahn, 2018), and long-term job trends suggest a greater emphasis on nonroutine tasks that require more critical thinking (Atalay et al., 2020). Further, these conclusions are corroborated by changes in observed returns to skills from surveys like the ACS, providing a convincing argument that the digital trace measures have construct validity.

Skill Development.

As discussed in the preceding section, advocates for the liberal arts argue that it develops economically valuable skills. Reflecting on the state of research on human capital, Deming (2022) concludes that soft skills – for example, critical thinking and teamwork – are increasingly economically valuable but the process of developing these skills is not well understood. While evidence from online job ads and long-running surveys suggest that the skills liberal arts education claims have value, it has been empirically difficult to link the practices of a liberal arts education that develop skills to labor market outcomes. Digital trace data offer two proxies for

skill development that can address these challenges: skill development inferred from syllabi and skill development inferred from textbooks,

Measuring Skill Development with Syllabi. First, consider the role syllabi can play in measuring skill development. Drawing on natural language processing techniques, Yu et al., (2021) offer a research framework for measuring the alignment of educational content and labor market outcomes, arguing the increasing digitization of text descriptions of course content through syllabi and online job boards provide an opportunity to develop microscopic measures of the alignment between curriculum and the labor market. Echoing Yu et al.'s research framework, Chau et al. (2023) map the text of syllabi to the O*Net hierarchical taxonomy of skills using natural language processing to measure the semantic similarity of sentences in the syllabi and the short textual descriptions of tasks described in O*Net. Drawing on the Open Syllabus Corpus, they measure the semantic similarity of 1.4 million syllabi at 3,000 college and universities in the United States, then link these measures with aggregated labor market outcomes from the College Scorecard. To derive measures connecting the text of syllabi to O*NET, they compare the semantic similarity of a syllabus's title, description, and learning objectives to the textual description of 2070 work tasks using fastText word embeddings trained on Wikipedia pages. For each task – for example “forecast economic, political, or social trends” – this provides a similarity score that measures how similar the language used in sentences in syllabi are to that of the task. They aggregate these scores to the field of study, allowing application of an agglomerative hierarchical clustering algorithm that shows how skills co-occur across disciplines. They show the aggregated field of study scores are useful predictors of average field by institution level wages one year after graduation and the likelihood that a work activity will become important in the future.

While their measures of semantic similarity provide a strong starting point to measure skills taught in syllabi at scale, Chau et al. focus primarily on prediction, rather than developing and validating measures that can be used to explain social scientific phenomena. Without an argument that convincingly argues for the construct validity of these measures, we can conclude that they are useful for predicting skill change and wages but must remain skeptical about whether they measure meaningful theoretical constructs.

Measuring Skill Development with Assigned Textbooks. Particularly in the hard sciences, mathematics, and engineering, the content of textbooks can also be a proxy for the skills students acquire. Researchers interested in how quantitative reasoning skills are incorporated throughout the curriculum, for example, might use the content of textbooks to identify courses outside of math departments that require mathematical content. For skills that clearly map onto the content of a book, digital trace data from textbooks then provide a measure of exposure to skill development. Assigning a textbook on the SQL database programming language likely indicates a course teaches SQL. In contrast, softer skills like writing cannot be as easily inferred from assigned textbooks. While courses with textbooks about writing likely teach writing, many courses in philosophy, history, and English that do not assign textbooks with explicit references to writing do so as well.

Pedagogical Practices.

Measuring Pedagogical Practices with Lecture Capture Recordings. Liberal arts classrooms have been associated with a greater use of active learning activities such as small-group discussions and activities (Pascaeralla et al., 2005). Data on these claims is typically drawn from self-reported surveys of faculty teaching practices and student perceptions of classrooms or - less frequently – labor-intensive qualitative observation of classrooms

(Campbell, 2023). Drawing on qualitative observations of 700 courses, Campbell (2023) compares the teaching practices of five liberal arts colleges to two regional public institutions, a public research university, and an elite private college. Courses in liberal arts colleges were more likely to incorporate higher rigor, required greater cognitive complexity, placed greater expectations on student preparation for course and participation, and were more likely to incorporate effective teaching strategies including instructors asking students questions in class, having students ask questions, and incorporating visuals and activities.

As classrooms are now equipped with lecture capture technology and some courses are taught solely online, there has been an explosion of video data capturing the classroom environment. Machine learning methods applied to video and text transcripts from lecture capture systems provide digital traces that can measure the dynamics of classrooms at scale. Although measures derived from video observation using machine learning will be less accurate than qualitative observation, they serve as a complement, sacrificing some accuracy for scale and the ability to be always-on. For instance, an emerging field of research in the K-12 context uses the transcripts from video recordings of classrooms to measure teaching practices at scale (for example, Wang & Demszky, 2023; Demszky, Liu, Hill, Sanghi, & Chung, 2023; Demszky Liu, Hill, Jurafsky, & Piech, 2023; Liu & Cohen, 2021, Xu et al., 2024).

Unlike many forms of digital trace data, however, lecture capture may not benefit from nonreactivity. Given that political activists have encouraged students to record faculty they disagree with (Kolowich and Quintana, 2017), instructors that know they are being recorded may change their pedagogical practices and the content they are covering or opt out of participating in lecture capture. In the former case, our measure of faculty teaching becomes reactive; in the latter, our measure becomes systematically incomplete.

Student Academic Engagement.

Measuring Student Academic Engagement with Learning Management Systems.

Liberal arts colleges are associated with greater student engagement in educationally purposeful activities (Pascarella et al., 2005). However, quantitative measures of student engagement are typically created through student self-reports like the National Survey of Student Engagement. Website clickstream data provide an alternative means to measuring student academic engagement. Clickstreams are timestamped events that are passively recorded as students interact with learning management systems, providing fine-grained time-variant information that tracks student learning (Baker et al., 2020). Researchers can derive partial measures about student engagement with course content from clickstream data from LMS systems; for example, Park et al. (2017) construct daily measures of student interactions with course material where an activity involved previewing or reviewing course materials. They find patterns of interaction with course materials was significantly associated with a student's final course grade. Because clickstreams records are always-on passively recording student's behavior, these measures are timelier and – in some ways – more objective than self-reports of student engagement (Baker et al. 2020).

However, a student that does not interact with course materials on a learning management system may be engaging with the course material in other ways (Baker et al., 2020). A student could be watching videos with a friend or could access the class readings through other websites. As a result, clickstreams are noisy and incomplete measures of student engagement. Further, faculty may use learning management systems in different ways. A faculty member that requires students to participate in online discussion boards will cause more students to access a learning management system than one that does not.

Social Interaction.

A liberal arts education has historically been associated with residential campuses and engagement across diverse perspectives (Detweiler, 2021; Pascarella et al., 2005). This is a distinctively American phenomena that is uncommon in most postsecondary institutions worldwide (Delbanco, 2012), raising questions about its value. Theoretically, living on campus might increase student involvement on campus (Astin & Antonio, 2012) and may provide students with access to resources and information about college through more regular interaction with peers and university staff (Schudde, 2011). However, simply living on campus would be insufficient to generate the theoretical benefits: a student could simply stay in their room, and never interact with others. To demonstrate the relationship between living on campus and outcomes, we require measures of interpersonal interactions, the assumed mechanism that links residential living to outcomes. In this respect, understanding the processes through which a liberal arts education requires measures of the networked and relational lives of students, a task that Macy (2016) notes digital trace data are especially apt for measuring. Leveraging digital trace data can examine the distinctively American practice of residential colleges, allowing social scientists to measure what Stevens et al. (2008) call the “network dynamics at the experiential core of college” (p. 134).

Measuring Social Interactions with Email. A liberal arts education has historically been characterized by meaningful interactions with faculty (Detweiler, 2021). A natural proxy for student-faculty interaction comes in the form of email between students and faculty (Nelson Laird et al., 2008). While critics can rightfully argue that email is a limited measure of faculty interaction, experimental evidence suggests that two or three carefully timed emails from faculty can causally increase student satisfaction, grades, credit accumulation, and degree attainment among structurally disadvantaged student populations (Carrell & Kurlaender, 2023).

Kossinets and Watts (2006) construct a yearlong longitudinal network of email exchanges comprised of 43,553 students, staff, and faculty at a large university. Their network is comprised of nodes of individuals connected by edges defined by emails; for a given point in time, two individuals share an edge if they have emailed at any point in the previous 60 days. They find that students are more likely to share edges if they enroll in the same class and show that there is not meaningful homophily across demographics facets like gender and age. This suggests that email can be captured at scale and linked with student demographics to understand how student-faculty interactions occur in this medium. If liberal arts education is characterized by more meaningful interaction with faculty, we might anticipate that there will be more email exchanges between students and faculty at liberal arts colleges that can be measured in these digital traces.

Measuring Social Interactions with WiFi Router Logs. WiFi routers can provide always-on digital traces of student interactions. Drawing on WiFi Router signals, Nguyen et al. (2020) link academic records with proxies of student locations, facilitating such a measure. When accessing university WiFi, student devices connect to routers throughout campus that have limited ranges. By identifying the digital devices each student has registered with the university and the routers they connect to at each time point, Nguyen et al. are able to repurpose WiFi access logs to provide fine-grained proxies of student locations. They then construct social networks of students based on co-occurrences near the same WiFi routers. Nguyen et al. (2020) use the resulting social networks to test whether student networks are characterized by homophily – showing that students are more likely to interact with other students who are similar on the basis of academic achievement, gender, race, and family income. Building on this work, repurposing digital trace measures of WiFi routers as measures of student’s social networks

could provide longitudinal and fine-grained measures of student, staff, and faculty social networks on campus. However, using routers in this way assumes that most students, staff, and faculty use devices that connect to WiFi routers and have those devices with them much of the time. Any individuals that do not connect to WiFi routers are ignored. Further, WiFi routers will only capture interactions on campus, although significant student interactions at non-residential colleges likely occurs in off-campus student housing. While Nguyen et al. provide some evidence that the social networks inferred through this digital trace data is predictive of student grades, they do not validate the resulting social network data through other means.

Measuring Social Interactions with Social Media. Overgoor et al. (2020) use Facebook data to derive the online social networks of students from 1,159 institutions, linking these networks with data from the College Scorecard. In their networks, each student is a node and students are connected if they are friends. They find that students at private schools have more connections to other students at their school and the networks of students at larger schools are less dense, suggesting greater connectedness and interaction at private institutions.

In the Copenhagen Network Study, Stopczynski et al. (2014) demonstrate the feasibility of combining several distinct digital traces of social interactions. They measure 1,000 students' social networks by combining data from WiFi routers, email exchanges, social media data from Facebook, traditional social science surveys, a custom Bluetooth app that detected when two phones were in close proximity to one another, location data from phone sensors, text and call records from phones, and an embedded anthropologist who provided complimentary qualitative analysis of the developing social network. Arguing that relying on a single measure provides an incomplete picture of a student's true social network, they show how these digital trace data captures distinct channels through which social interactions occur. In their data, any one single

stream of digital trace data produces an inaccurate measure of a student's true social network: measuring a student's social network through physical proximity inferred by their Bluetooth app missed 23.6 percent of a student's Facebook network.

V. Digitally Tracing the Liberal Arts

Digital trace data offer many opportunities to measure the processes and outcomes of a liberal arts education. Table 2 summarizes the various sources of digital trace data available to do so. However, because each digital trace is a partial and incomplete measure of a student's educational experience, much of the power of digital trace data in the social sciences comes from linking across several kinds of data (Lazer & Radford, 2017). Data linkages between digital trace data and other forms of data allow for long-term follow-ups on experiments and the ability to supplement survey data with a wealth of additional information from digital traces (Lazer & Radford, 2017; Salganik, 2018). This include linking digital trace data to surveys, linking to other forms digital trace data, and using large scale digital trace data to identify interesting cases for qualitative investigations (Lazer & Radford, 2017; Salganik, 2018). Because digital trace data often span multiple levels of educational processes these data linkages can allow researchers to examine the mechanisms through which a liberal arts education operates: for instance, by combining extremely granular student learning behavior traced in clickstream logs, mesolevel data about the assignments students are engaging in through syllabi, and macro level data on student majors and labor market outcomes. Combining several sources of digital trace data in this way can link theories of micro and macro level processes, allowing for richer theorizer (Edelmann et al., 2020).

Table 2

Digital Trace Measures of Liberal Arts Education Processes and Outcomes

Theoretical Construct	Digital Trace Data
-----------------------	--------------------

Course Expectations	Syllabi Learning Management Systems
Course Content	Course Titles from Transcripts Assigned Textbooks
Demand for Skills	Online Job Advertisements Digitized Newspapers
Skill Development	Syllabi Assigned Textbooks
Pedagogical Practices	Lecture Capture
Student Academic Engagement	Website Clickstreams
Social Interaction and Social Networks	WiFi Routers Emails Social Media Networks

To illustrate the potential of combining digital traces, consider the following research question: what are the economic consequences of a liberal arts education's emphasis on student engagement? Drawing on digital trace data, we can derive quantitative measures of factors associated with student engagement that measure engagement at multiple theoretical levels. Digital traces from emails serve as a proxy for student-faculty engagement (Carell & Kurlaender, 2023) and we can infer whether students interact with faculty based on connections to the same WiFi routers (Stopczynski et al., 2014; Nguyen et al., 2020). The social interactions of students on a residential campus can be proxied through social media websites like Facebook (Stopczynski, 2014; Overgoor et al, 2020), connections to WiFi routers (Stopczynski et al., 2014; Nguyen et al., 2020), and email (Kossinets and Watts, 2006; Stopczynski et al., 2014). Proxies for high impact

practices theorized to increase student engagement are passively recorded in a number of data sources: internships and study abroad are captured in the titles of courses in administrative transcripts (Paulson et al., 2023, Zhou, 2023), group projects and writing assignments can be observed in the text of syllabi (Biasi & Ma, 2022), and submitted essays can be extracted from learning management systems (Lewis et al., 2005; Fischer et al., 2020). We can observe micro-level digital traces of student academic engagement in the clickstreams of students interacting with learning management systems (Baker et al., 2020) and can capture practices of active learning in the classroom from video and text transcripts in lecture capture systems (Liu & Cohen, 2021). Finally, we can observe the long-term labor market outcomes of students through administrative Unemployment Insurance and tax records. Each digital trace provides a partial and incomplete picture of student engagement and its consequences; together they paint a detailed portrait of student engagement that can be linked with long-term economic outcomes and supplemented with traditional social science surveys and qualitative interviews to validate the measures.

However, the theoretical utility of combining multiple kinds of digital trace data runs into what Lazer and Radford (2017) calls the challenge of “too many big data”: theoretically linked behaviors are distributed across many datasets, each of which is governed by a different authority. Linking many sources of digital trace data results in practical and technical challenges due to their incomplete, inaccessible, and sensitive nature. Challenges to accessing fragmented sources of digital trace data can make research prohibitively expensive if there is not a robust research infrastructure to support it (Penner & Dodge, 2019). Because these data are not typically designed for research purposes, accessing each additional dataset may require convincing a new stakeholder of the scholarly value of the research (and often the practical benefit to their

organization). Given that stakeholders that can grant access to digital trace data like email exchanges and WiFi routers are seldom social scientists, there may be additional challenges translating the rationale for the data request across disciplinary boundaries. Even when data access is granted, linkage requires shared identifiers across distinct datasets. Because they are not designed for research purposes, digital trace data like clickstreams may lack the necessary variables to perform merges across different kinds of digital trace data (Baker et al., 2020). For instance, email exchanges may be organized by username while administrative transcripts contain only have student ID numbers and syllabi might be labeled by internal course registration numbers rather than section numbers.

Surmounting the challenges with linking and using digital trace data for social science research requires a substantial investment in research infrastructure before we can effectively leverage the always-on passive collection of data in digital traces to understand educational processes. Just as the credibility revolution was driven by a substantial investment in data infrastructure to facilitate research with administrative data (Card et al., 2010; Figlio et al., 2017; Penner & Dodge, 2019), research with other forms of digital trace data will require investment by institutions in data infrastructure and staff familiar with the principles of computational social science to make it possible for researchers to securely access and link multiple forms of digital trace data.

Beyond data infrastructure, successfully leveraging the tools of computational social science to quantitatively study the value of a liberal arts education requires interdisciplinary teams. Using computational social science to student the liberal arts – perhaps fittingly - requires computer scientists curious about the liberal arts who have the technical skills required to access and clean big dirty data, social scientists who can recognize how the digital transformation of

educational processes provide access to new measures of theoretical constructs, and researchers familiar with the philosophical arguments and historical tradition that shaped contemporary liberal arts education who are comfortable translating abstract constructs into quantitative measures. Although the digital transformation of educational processes means social scientists can now measure key theoretical constructs related to liberal arts education through digital traces, forming a research agenda around this data will require robust research infrastructure and interdisciplinary teams to navigate the challenges and affordances of digital trace data.

References

- Amornsiripanitch, N., Gompers, P., Hu, G., Levinson, W., & Mukharlyamov, V. (2023). *No Revenge for Nerds? Evaluating the Careers of Ivy League Athletes* (Working Paper 31753). National Bureau of Economic Research. <https://doi.org/10.3386/w31753>
- Andrews, R. J., Imberman, S. A., Lovenheim, M. F., & Stange, K. M. (2022). The Returns to College Major Choice: Average and Distributional Effects, Career Trajectories, and Earnings Variability. *National Bureau of Economic Research Working Paper Series*.
- Arum, R., & Roksa, J. (2011). *Academically adrift: Limited learning on college campuses*. University of Chicago Press.
- Astin, A. (1993). *What Matters in College: Four Critical Years Revisited*. San Francisco: Jossey-Bass.
- Astin, A. W., & Antonio, Anthony. (2012). *Assessment for excellence: The philosophy and practice of assessment and evaluation in higher education*. Rowman & Littlefield Publishers.
- Atalay, E., Phongthientham, P., Sotelo, S., & Tannenbaum, D. (2020). The Evolution of Work in the United States. *American Economic Journal: Applied Economics*, 12(2), 1–34. <https://doi.org/10.1257/app.20190070>
- Baker, R., Xu, D., Park, J., Yu, R., Li, Q., Cung, B., Fischer, C., Rodriguez, F., Warschauer, M., & Smyth, P. (2020). The benefits and caveats of using clickstream data to understand student self-regulatory behaviors: Opening the black box of learning processes. *International Journal of Educational Technology in Higher Education*, 17(1), 13. <https://doi.org/10.1186/s41239-020-00187-1>

- Biasi, B., & Ma, S. (2022). *The education-innovation gap*. National Bureau of Economic Research.
https://www.nber.org/system/files/working_papers/w29853/revisions/w29853.rev1.pdf
- Bok, D. (2006). *Our Underachieving Colleges: A Candid Look at How Much Students Learn and Why They Should Be Learning More*. Princeton University Press.
- Borisova, I. (2018). College Course Name Classification at Scale. In C. Penstein Rosé, R. Martínez-Maldonado, H. U. Hoppe, R. Luckin, M. Mavrikis, K. Porayska-Pomsta, B. McLaren, & B. du Boulay (Eds.), *Artificial Intelligence in Education* (pp. 419–423). Springer International Publishing. https://doi.org/10.1007/978-3-319-93846-2_78
- Börner, K., Scrivner, O., Gallant, M., Ma, S., Liu, X., Chewning, K., Wu, L., & Evans, J. A. (2018). Skill discrepancies between research, education, and jobs reveal the critical need to supply soft skills for the data economy. *Proceedings of the National Academy of Sciences of the United States of America*, 115(50), 12630–12637.
<https://doi.org/10.1073/pnas.1804247115>
- Brighouse, H. (2019). *How can we understand ‘liberal arts education’?* Andrew W. Mellon Foundation.
- Campbell, C. (2023). *Great College Teaching: Where it Happens and how to Foster it Everywhere*. Harvard Education Press.
- Card, D. E., Chetty, R., Feldstein, M. S., & Saez, E. (2010). Expanding Access to Administrative Data for Research in the United States. *SSRN Electronic Journal*.
<https://doi.org/10.2139/ssrn.1888586>

- Carrell, S. E., & Kurlaender, M. (2023). My Professor Cares: Experimental Evidence on the Role of Faculty Engagement. *American Economic Journal: Economic Policy*, 15(4), 113–141. <https://doi.org/10.1257/pol.20210699>
- Chau, H., Bana, S. H., Bouvier, B., & Frank, M. R. (2023). Connecting higher education to workplace activities and earnings. *PLOS ONE*, 18(3), e0282323. <https://doi.org/10.1371/journal.pone.0282323>
- Conzelmann, J. G., Hemelt, S. W., Hershbein, B. J., Martin, S., Simon, A., & Stange, K. M. (2023). Grads on the go: Measuring college-specific labor markets for graduates. *Journal of Policy Analysis and Management*, n/a(n/a). <https://doi.org/10.1002/pam.22553>
- Cooper, A. (2022) Educational Programs. In S. Moody-Turner (Ed.). *The Portable Anna Julia Cooper*. Penguin Books. 190-204. (Original work published 1930.)
- Crossley, S., Wan, Q., Allen, L., & McNamara, D. (2023). Source inclusion in synthesis writing: An NLP approach to understanding argumentation, sourcing, and essay quality. *Reading and Writing*, 36(4), 1053–1083. <https://doi.org/10.1007/s11145-021-10221-x>
- Currie, J., Kleven, H., & Zwiers, E. (2020). Technology and Big Data Are Changing Economics: Mining Text to Track Methods. *AEA Papers and Proceedings*, 110, 42–48. <https://doi.org/10.1257/pandp.20201058>
- Delbanco, A. (2012). *College: What it was, is, and should be*. Princeton University Press.
- Deming, D. J. (2017). The Growing Importance of Social Skills in the Labor Market. *The Quarterly Journal of Economics*, 132(4), 1593–1640. <https://doi.org/10.1093/qje/qjx022>
- Deming, D. J. (2022). Four Facts about Human Capital. *Journal of Economic Perspectives*, 36(3), 75–102. <https://doi.org/10.1257/jep.36.3.75>

- Deming, D. J., & Noray, K. (2020). Earnings Dynamics, Changing Job Skills, and STEM Careers. *The Quarterly Journal of Economics*, 135(4), 1965–2005.
<https://doi.org/10.1093/qje/qjaa021>
- Deming, D., & Kahn, L. B. (2018). Skill Requirements across Firms and Labor Markets: Evidence from Job Postings for Professionals. *Journal of Labor Economics*, 36, S337–S369. <https://doi.org/10.1086/694106>
- Demszky, D., Liu, J, Hill, H., Sanghi, S., & Chung, A. (2023). *Improving Teachers' Questioning Quality through Automated Feedback: A Mixed-Methods Randomized Controlled Trial in Brick-and-Mortar Classrooms* (EdWorkingPaper: 23-875). Annenberg Institute at Brown University. <https://doi.org/10.26300/8PNW-5Q67>
- Demszky, D., Liu, J., Hill, H., Jurafsky, D., & Piech, C. (2023). Can Automated Feedback Improve Teachers' Uptake of Student Ideas? Evidence From a Randomized Controlled Trial in a Large-Scale Online Course. *Educational Evaluation and Policy Analysis*.
<https://doi.org/10.3102/01623737231169270>
- Detweiler, R. (2021). *The Evidence the Liberal Arts Needs*. MIT Press.
- D'Ignazio, C., & Klein, L. F. (2020). *Data feminism*. MIT press. <https://direct.mit.edu/books/oa-monograph/4660/Data-Feminism>
- Du Bois, W. E. B. (2004). The Talented Tenth. In Zuckerman, P. (Ed.). *The Social Thought of W.E.B. Du Bois*. SAGE Publications, Inc. 185-206. (Original work published in 1903).
- Du Bois, W. E. B. (2004). The Negro College. In Zuckerman, P. (Ed.). *The Social Thought of W.E.B. Du Bois*. SAGE Publications, Inc. 185-206. (Original work published in 1933).

- Edelmann, A., Wolff, T., Montagne, D., & Bail, C. A. (2020). Computational Social Science and Sociology. *Annual Review of Sociology*, 46(1), 61–81. <https://doi.org/10.1146/annurev-soc-121919-054621>
- Figlio, D., Karbownik, K., & Salvanes, K. (2017). The Promise of Administrative Data in Education Research. *Education Finance and Policy*, 12(2), 129–136. https://doi.org/10.1162/EDFP_a_00229
- Fischer, C., Pardos, Z. A., Baker, R. S., Williams, J. J., Smyth, P., Yu, R., Slater, S., Baker, R., & Warschauer, M. (2020). Mining Big Data in Education: Affordances and Challenges. *Review of Research in Education*, 44(1), 130–160. <https://doi.org/10.3102/0091732X20903304>
- Foote, A., & Stange, K. M. (2022). *Attrition from Administrative Data: Problems and Solutions with an Application to Postsecondary Education* (Working Paper 30232). National Bureau of Economic Research. <http://www.nber.org/papers/w30232>
- Goldsmith-Pinkham, P. (2024). Tracking the Credibility Revolution across Fields (arXiv:2405.20604). arXiv. <http://arxiv.org/abs/2405.20604>
- Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). *Text as data: A new framework for machine learning and the social sciences*. Princeton University Press.
- Hart Research Associates. (2016). *Recent Trends in General Education Design, Learning Outcomes, and Teaching Approaches*. <https://files-eric-ed.gov.proxy.lib.umich.edu/fulltext/ED582012.pdf>
- Hemelt, S. W., Hershbein, B., Martin, S., & Stange, K. M. (2023). College majors and skills: Evidence from the universe of online job ads. *Labour Economics*, 85, 102429. <https://doi.org/10.1016/j.labeco.2023.102429>

Higher Education Opportunity Act, House Resolution 4137, 110th Congress (2008).

<https://www.congress.gov/bill/110th-congress/house-bill/4137>

HIQ Labs, Inc. v. LinkedIn Corporation, No. 17-16783 (9th Cir. 2022).

Jiang, W., & Pardos, Z. A. (2020). Evaluating Sources of Course Information and Models of Representation on a Variety of Institutional Prediction Tasks. *International Educational Data Mining Society*.

Kabo, F., Paulson, A., Bradley, D., Varnum, K., & Teasley, S. (2024). Longitudinal Associations between Online Resource Usage and Undergraduate Student Performance. *College & Research Libraries*, forthcoming.

Kilgo, C. A., Ezell Sheets, J. K., & Pascarella, E. T. (2015). The link between high-impact practices and student learning: Some longitudinal evidence. *Higher Education*, 69(4), 509–525. <https://doi.org/10.1007/s10734-014-9788-z>

Kolowich, S., & Quintana, C. (2017, December 21). A Brief History of Students Secretly Recording Their Professors. *The Chronicle of Higher Education*.
<https://www.chronicle.com/article/a-brief-history-of-students-secretly-recording-their-professors/>

Kossinets, G., & Watts, D. J. (2006). Empirical analysis of an evolving social network. *Science*, 311(5757), 88–90.

Kuh, G. D. (2008). *High-impact educational practices: What they are, who has access to them, and why they matter*. Association of American Colleges and Universities.

Lazer, D., Kennedy, R., King, G., & Vespignani, A. (2014). The Parable of Google Flu: Traps in Big Data Analysis. *Science*, 343(6176), 1203–1205.
<https://doi.org/10.1126/science.1248506>

- Lazer, D., & Radford, J. (2017). Data ex Machina: Introduction to Big Data. *Annual Review of Sociology*, 43(1), 19–39. <https://doi.org/10.1146/annurev-soc-060116-053457>
- Lewis, B., MacEntee, V., DeLaCruz, S., Englander, C., Jeffrey, T., Takach, E., Wilson, S., & Woodall, J. (2005). *Learning Management Systems Comparison*. InSITE 2005: Informing Science + IT Education Conference. <https://doi.org/10.28945/2873>
- Liu, J., & Cohen, J. (2021). Measuring Teaching Practices at Scale: A Novel Application of Text-as-Data Methods. *Educational Evaluation and Policy Analysis*, 43(4), 587–614. <https://doi.org/10.3102/01623737211009267>
- Macy, M. W. (2016). An Emerging Trend: Is Big Data the End of Theory? In R. A. Scott & S. M. Kosslyn (Eds.), *Emerging Trends in the Social and Behavioral Sciences* (1st ed., pp. 1–14). Wiley. <https://doi.org/10.1002/9781118900772.etrds0410>
- Margolin, D., Lin, Y.-R., Brewer, D., & Lazer, D. (2013). Matching Data and Interpretation: Towards a Rosetta Stone Joining Behavioral and Survey Data. *Proceedings of the International AAAI Conference on Web and Social Media*, 7(5), Article 5. <https://doi.org/10.1609/icwsm.v7i5.14490>
- Montás, R. (2021). *Rescuing Socrates: How the great books changed my life and why they matter for a new generation*. Princeton University Press.
- National Leadership Council for Liberal Education and America's Promise. (2007). *College learning for the new global century: A report from the National Leadership Council for Liberal Education & America's Promise*. Association of American Colleges and Universities.

- Nelson Laird, T. F. N., Chen, D., & Kuh, G. D. (2008). Classroom practices at institutions with higher-than-expected persistence rates: What student engagement data tell us. *New Directions for Teaching and Learning*, 2008(115), 85–99. <https://doi.org/10.1002/tl.327>
- Nelson Laird, T. F., Niskodé-Dossett, A. S., & Kuh, G. D. (2009). What General Education Courses Contribute to Essential Learning Outcomes. *The Journal of General Education*, 58(2), 65–84.
- Nguyen, Q., Poquet, O., Brooks, C., & Li, W. (2020). Exploring Homophily in Demographics and Academic Performance Using Spatial-Temporal Student Networks. *International Educational Data Mining Society*.
- Open Syllabus Project. (n.d.). *Open Syllabus Explorer*. <https://explorer.opensyllabus.org/>
- Overgoor, J., State, B., & Adamic, L. (2020). *The Structure of U.S. College Networks on Facebook* (arXiv:2006.04980). arXiv. <https://doi.org/10.48550/arXiv.2006.04980>
- Park, J., Denaro, K., Rodriguez, F., Smyth, P., & Warschauer, M. (2017). Detecting changes in student behavior from clickstream data. *Proceedings of the Seventh International Learning Analytics & Knowledge Conference*, 21–30. <https://doi.org/10.1145/3027385.3027430>
- Pascarella, E. T., Wolniak, G. C., Seifert, T. A., Cruce, T. M., & Blaich, C. F. (2005). Liberal Arts Colleges and Liberal Arts Education: New Evidence on Impacts. ASHE Higher Education Report, Volume 31, Number 3. *ASHE Higher Education Report*, 31(3), 1–148.
- Paulson, A., Stange, K., & Flaster, A. (2024). *Classifying Courses at Scale: A Text as Data Approach to Characterizing Student Course-Taking Trends with Administrative Transcripts*. (EdWorkingPaper: 24-1042). Annenberg Institute at Brown University. . <https://edworkingpapers.com/ai24-1042>

- Penner, A. M., & Dodge, K. A. (2019). Using Administrative Data for Social Science and Policy. *The Russell Sage Foundation Journal of the Social Sciences*, 5(3), 1–18.
<https://doi.org/10.7758/RSF.2019.5.3.01>
- Rajkumar, K., Saint-Jacques, G., Bojinov, I., Brynjolfsson, E., & Aral, S. (2022). A causal test of the strength of weak ties. *Science*, 377(6612), 1304–1310.
<https://doi.org/10.1126/science.abl4476>
- Rivero, A. C., Samuel Bestvater, Emma Remy and Gonzalo. (2024, May 17). When Online Content Disappears. *Pew Research Center*. <https://www.pewresearch.org/data-labs/2024/05/17/when-online-content-disappears/>
- Salganik, M. J. (2018). *Bit by bit: Social research in the digital age*. Princeton University Press.
- Sasse, B. (2023, December 14). The Moral Decline of Elite Universities. *The Atlantic*.
<https://www.theatlantic.com/ideas/archive/2023/12/intersectionality-american-college/676350/>
- Schudde, L. T. (2011). The causal effect of campus residency on college student retention. *The Review of Higher Education*, 34(4), 581–610.
- Stanley, J. (2020). *How fascism works: The politics of us and them*. Random House Trade Paperbacks.
- Stevens, M. L., Armstrong, E. A., & Arum, R. (2008). Sieve, Incubator, Temple, Hub: Empirical and Theoretical Advances in the Sociology of Higher Education. *Annual Review of Sociology*, 34(1), 127–151. <https://doi.org/10.1146/annurev.soc.34.040507.134737>
- Stopczynski, A., Sekara, V., Sapiezynski, P., Cuttone, A., Madsen, M. M., Larsen, J. E., & Lehmann, S. (2014). Measuring Large-Scale Social Networks with High Resolution. *PLoS ONE*, 9(4), e95978. <https://doi.org/10.1371/journal.pone.0095978>

- Wang, R., & Demszky, D. (2023). Is ChatGPT a Good Teacher Coach? Measuring Zero-Shot Performance For Scoring and Providing Actionable Insights on Classroom Instruction. In E. Kochmar, J. Burstein, A. Horbach, R. Laarmann-Quante, N. Madnani, A. Tack, V. Yaneva, Z. Yuan, & T. Zesch (Eds.), *Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2023)* (pp. 626–667). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2023.bea-1.53>
- Wells, C. A. (2016). Realizing General Education: Reconsidering Conceptions and Renewing Practice. *ASHE Higher Education Report*, 42(2), 1–85.
<https://doi.org/10.1002/aehe.20068>
- Wildavsky, B. (2023). *The Career Arts: Making the Most of College, Credentials and Connections*. Princeton University Press.
- Xu, P., Liu, J., Jones, N., Cohen, J., & Ai, W. (2024). The Promises and Pitfalls of Using Language Models to Measure Instruction Quality in Education. *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 4375–4389.
<https://doi.org/10.18653/v1/2024.naacl-long.246>
- Yu, R., Das, S., Gurajada, S., Varshney, K., Raghavan, H., & Lastra-Anadon, C. (2021). A Research Framework for Understanding Education-Occupation Alignment with NLP Techniques. *Proceedings of the 1st Workshop on NLP for Positive Impact*, 100–106.
<https://doi.org/10.18653/v1/2021.nlp4posimpact-1.11>
- Zhou, R. Y. (2023). *Understanding Experiential Learning Through Work-Based College Coursetaking: Evidence From Transcript Data Using a Text Mining Technique* [CCRC

Working Paper No. 131]. <https://ccrc.tc.columbia.edu/media/k2/attachments/experiential-learning-work-based-coursetaking.pdf>